



Using the Telnetlib Module to Configure Switches

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives	2
2 Task Preparation	2
2.1 Network Topology	2
2.2 Initial Configuration	2
3 Task Implementation	5
4 Verification	12
5 Quiz	12



1 About This Lab

1.1 Task Description

The telnetlib module is a standard Python module used for remote Telnet-based management of network devices. It enables automated connections to devices and execution of corresponding operations.

After completing this task, you will have a basic understanding of network automation and know how to access and configure devices using telnetlib in Python.

1.2 Objectives

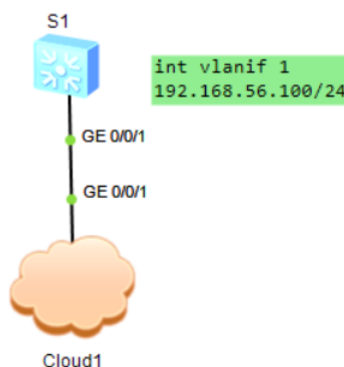
On completing this task, you will be able to:

1. Learn the basic Python syntax.
2. Learn how to use telnetlib.

2 Task Preparation

2.1 Network Topology

Figure 1 Using the telnetlib module to configure a switch



2.2 Initial Configuration

- The initial configuration of S1 is as follows.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]int vlanif1
[S1-Vlanif1]ip address 192.168.56.100 24
[S1-Vlanif1]quit
[S1]
```



- Configure S1 as a Telnet server.

```
[S1]telnet server enable
Info: The Telnet server has been enabled.
[S1]aaa
[S1-aaa]local-user user01 password cipher Huawei12#$
Info: Add a new user.
[S1-aaa]local-user user01 service-type telnet
[S1-aaa]quit
[S1]user-interface vty 0 4
[S1-ui-vty0-4]authentication-mode aaa
[S1-ui-vty0-4]user privilege level 3
[S1-ui-vty0-4]protocol inbound telnet
[S1-ui-vty0-4]quit
[S1]
```

- Perform a telnet operation on S1 to test its own IP address.

```
<S1>telnet 192.168.56.100
Trying 192.168.56.100 ...
Press CTRL+K to abort
Connected to 192.168.56.100 ...

Login authentication

Username:user01
Password:Huawei12#$
Apr  5 2025 22:33:50-08:00 S1 %%01SHELL/4/LOGINFAILED(I)[1]:Failed to login. (Ip
=192.168.56.100, UserName=user01, Times=1, AccessType=TELNET)
Error: Local authentication is rejected

Username:user01
Password:
Info: The max number of VTY users is 5, and the number
      of current VTY users on line is 1.
      The current login time is 2025-04-05 22:34:07.
<S1>
<S1>quit
Info: The max number of VTY users is 5, and the number
      of current VTY users on line is 0.
Info: The connection was closed by the remote host.
<S1>
```



- Configure the cloud in the topology to connect it to the local PC, as shown in Figure 2.

Figure 2 Connecting the cloud to the local PC

The screenshot shows the 'Cloud1' configuration window with the 'IO 配置' (IO Configuration) tab selected. The '端口创建' (Port Creation) section is active, displaying a table of created ports and a '端口映射设置' (Port Mapping Settings) section.

端口创建 (Port Creation) Section:

- 绑定信息:** VirtualBox Host-Only Network -- IP: 192.168.56.1
- 警告:** 请勿绑定公网网卡, 否则可能会引起网络瘫痪。
- 端口类型:** GE
- 监听端口:** 30000
- 对端IP:** 0 . 0 . 0 . 0
- 对端口:** 0
- 建议:** (30000-35000)
- 增加** and **删除** buttons are present.

端口列表 (Port List Table):

No.	端口类型	端口编号	UDP端口号	端口开放状态	绑定信息
1	GE	1	3857	Internal	UDP
2	GE	2	None	Public	VirtualBox Host-Only Network -- IP: 192.168.56.1

端口映射设置 (Port Mapping Settings) Section:

- 端口类型:** GE
- 入端口编号:** 1
- 出端口编号:** 2
- 双向通道:** ☒
- 增加** button is present.

端口映射表 (Port Mapping Table):

No.	入端口编号	出端口编号	端口类型
1	1	2	GE
2	2	1	GE

删除 button is present.

- Test connectivity between the local NIC and S1, as shown in Figure 3.



Figure 3 Testing connectivity between the local NIC and S1

```
管理员: C:\Windows\system32\cmd.exe
Microsoft Windows [版本 10.0.19045.5608]
(c) Microsoft Corporation。保留所有权利。

C:\Users\Administrator>ping 192.168.56.100 -t

正在 Ping 192.168.56.100 具有 32 字节的数据:
来自 192.168.56.100 的回复: 字节=32 时间=32ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=24ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=6ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=14ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=24ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=5ms TTL=255
来自 192.168.56.100 的回复: 字节=32 时间=22ms TTL=255

192.168.56.100 的 Ping 统计信息:
    数据包: 已发送 = 7, 已接收 = 7, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 5ms, 最长 = 32ms, 平均 = 18ms
Control-C
^C
C:\Users\Administrator>
```

3 Task Implementation

1. Checking Basic Configurations
Omitted.
2. Compiling a Python Script

```
import telnetlib
import time

host = '192.168.56.100'
username = 'user01'
password = 'Huawei12#$'

tn = telnetlib.Telnet(host)

tn.read_until(b"Username:")
tn.write(username.encode("ascii")+ b"\n")

tn.read_until(b"Password:")
tn.write(password.encode('ascii') + b"\n")
```



```
# Send the command for canceling the pagination display to the device.
tn.write(b"screen-length 0 temporary\n")

tn.write(b'display cu \n')
time.sleep(1)
print(tn.read_very_eager().decode('ascii'))

tn.write(b"system-view"+ b"\n")
tn.write(b"vlan batch 2 3 10"+ b"\n")

tn.write(b"int g0/0/10"+ b"\n")
tn.write(b"port link-type access"+ b"\n")
tn.write(b"port default vlan 2"+ b"\n")

tn.write(b"int g0/0/11"+ b"\n")
tn.write(b"port link-type access"+ b"\n")
tn.write(b"port default vlan 3"+ b"\n")

tn.write(b"int g0/0/12"+ b"\n")
tn.write(b"port link-type access"+ b"\n")
tn.write(b"port default vlan 10"+ b"\n")

tn.write(b'display vlan \n')
time.sleep(1)
print(tn.read_very_eager().decode('ascii'))

tn.close()
```

3. Running the Python Script

Info: The max number of VTY users is 5, and the number of current VTY users on line is 1.

The current login time is 2025-04-05 23:01:05.

<S1>screen-length 0 temporary

Info: The configuration takes effect on the current user terminal interface only.

<S1>display cu

#

sysname S1

#

vlan batch 2 to 3 10

#

cluster enable



```
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
 local-user user01 password cipher ,a$6XCF+U`/7`Q#CV&M21Q!!
 local-user user01 service-type telnet
#
interface Vlanif1
 ip address 192.168.56.100 255.255.255.0
#
interface MEth0/0/1
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface GigabitEthernet0/0/3
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
interface GigabitEthernet0/0/6
#
interface GigabitEthernet0/0/7
#
interface GigabitEthernet0/0/8
#
```



```
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
interface GigabitEthernet0/0/11
#
interface GigabitEthernet0/0/12
#
interface GigabitEthernet0/0/13
#
interface GigabitEthernet0/0/14
#
interface GigabitEthernet0/0/15
#
interface GigabitEthernet0/0/16
#
interface GigabitEthernet0/0/17
#
interface GigabitEthernet0/0/18
#
interface GigabitEthernet0/0/19
#
interface GigabitEthernet0/0/20
#
interface GigabitEthernet0/0/21
#
interface GigabitEthernet0/0/22
#
interface GigabitEthernet0/0/23
#
interface GigabitEthernet0/0/24
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
 authentication-mode aaa
 user privilege level 3
#
return
<S1>
```




```
system-view
```

```
Enter system view, return user view with Ctrl+Z.
```

```
[S1]vlan batch 2 3 10
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S1]int g0/0/10
```

```
[S1-GigabitEthernet0/0/10]port link-type access
```

```
[S1-GigabitEthernet0/0/10]port default vlan 2
```

```
[S1-GigabitEthernet0/0/10]int g0/0/11
```

```
[S1-GigabitEthernet0/0/11]port link-type access
```

```
[S1-GigabitEthernet0/0/11]port default vlan 3
```

```
[S1-GigabitEthernet0/0/11]int g0/0/12
```

```
[S1-GigabitEthernet0/0/12]port link-type access
```

```
[S1-GigabitEthernet0/0/12]port default vlan 10
```

```
[S1-GigabitEthernet0/0/12]display vlan
```

```
The total number of vlans is : 4
```

```
-----  
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;  
MP: Vlan-mapping;      ST: Vlan-stacking;  
#: ProtocolTransparent-vlan;  *: Management-vlan;  
-----
```

```
VID  Type  Ports
```

```
-----  
1    common  UT:GE0/0/1(U)    GE0/0/2(D)    GE0/0/3(D)    GE0/0/4(D)  
                GE0/0/5(D)    GE0/0/6(D)    GE0/0/7(D)    GE0/0/8(D)  
                GE0/0/9(D)    GE0/0/13(D)   GE0/0/14(D)   GE0/0/15(D)  
                GE0/0/16(D)   GE0/0/17(D)   GE0/0/18(D)   GE0/0/19(D)  
                GE0/0/20(D)   GE0/0/21(D)   GE0/0/22(D)   GE0/0/23(D)  
                GE0/0/24(D)  
2    common  UT:GE0/0/10(D)  
3    common  UT:GE0/0/11(D)  
10   common  UT:GE0/0/12(D)
```

```
VID  Status  Property      MAC-LRN Statistics Description
```

```
-----  
1    enable  default      enable  disable  VLAN 0001  
2    enable  default      enable  disable  VLAN 0002  
3    enable  default      enable  disable  VLAN 0003  
10   enable  default      enable  disable  VLAN 0010
```

```
[S1-GigabitEthernet0/0/12]
```



4. Code Interpretation

(1) Importing modules

```
import telnetlib
import time
```

Import the telnetlib and time modules required by the code. The two modules are built-in modules of Python and do not need to be installed.

By default, Python executes code sequentially without any interval. When Telnet is used to send configuration commands to a switch, the response may be delayed or the command output may be incomplete. In this case, you can use the sleep method of the time module to introduce intentional pauses between commands.

(2) Logging in to a device

Invoke multiple methods of the Telnet class in telnetlib to log in to S1.

```
host = '192.168.56.100'
password = 'Huawei12#$'
tn = telnetlib.Telnet(host)
```

Create two variables: **host** and **password**, which are the login address and password of the device respectively, and are the same as those configured on the device. In this example, only the Telnet password is configured for login. Therefore, no user name is required.

telnetlib.Telnet() calls the Telnet() method in the telnetlib class. This method involves login parameters, including the IP address and port number. If the port number is not specified, port 23 is used by default.

In this example, tn = telnetlib.Telnet(host) indicates that you log in to the device at 192.168.56.100 and assign the value of telnetlib.Telnet(host) to tn.

```
tn.read_until(b"Username:")
tn.write(username.encode("ascii")+ b"\n")
tn.read_until(b"Password:")
tn.write(password.encode('ascii') + b"\n")
```

When you log in to the device at 192.168.56.100 through Telnet, the following information is displayed:

```
<S1>telnet 192.168.56.100
Trying 192.168.56.100 ...
Press CTRL+K to abort
Connected to 192.168.56.100 ...

Login authentication
Username:user01
Password:
Info: The max number of VTY users is 5, and the number
      of current VTY users on line is 1.
      The current login time is 2025-04-05 23:07:33.
<S1>
<S1>
```



Note that the program does not know the end of information to be read. Therefore, `read_until()` is used to indicate that the program should read until the specified string in parentheses.

In this example, `tn.read_until(b"Username:")` indicates that the data is read until the string **Username:** is encountered. The prefix **b** indicates a byte string, which is required by the function since telnetlib operates on bytes rather than Unicode strings in Python 3. For more details, refer to the official telnetlib documentation. Omitting **b** would result in an error.

`tn.write(username.encode("ascii")+ b"\n")` indicates that after detecting the **Username:** string, the program inputs the predefined user name using `write()`. This parameter was previously defined as the Telnet login credential.

`tn.read_until(b"Password:")` indicates that the data is read until the string **Password:** is displayed. The prefix **b** indicates a byte string, which is required by the function since telnetlib operates on bytes rather than Unicode strings in Python 3. For more details, refer to the official telnetlib documentation. Omitting **b** would result in an error. After detecting the **Password:** string, the program inputs the predefined password using `write()`. This parameter was previously defined as the Telnet login credential.

In this example, `tn.write(password.encode('ascii') + b"\n")` consists of two parts: `password.encode('ascii')` and `b"\n"`. `password.encode('ascii')` converts the string **Huawei12#\$** into an ASCII-encoded value. The `+` operator concatenates the ASCII-encoded password with the newline character `b"\n"`, which simulates pressing **Enter**. This line sends the password **Huawei12#\$** followed by a carriage return.

(3) Issuing configuration commands

After logging in to the device through Telnet, use the Python script to issue commands to the device.

```
tn.write(b'display cu \n')
```

Continue using `write()` to issue commands to the device. The command **display cu** is short for **display current-configuration**, which is used to display the device's current configuration.

```
time.sleep(1)
```

`time.sleep(1)` pauses the program for one second. This delay allows the switch to return the command output before the next line of code runs. Without this delay, the subsequent code may execute before the output is available, resulting in unreadable or missing data.

```
print(tn.read_very_eager().decode('ascii'))
```

`print()` indicates that the contents in the parentheses are displayed on the console.

`tn.read_very_eager()` indicates reading as much data as possible.

`. decode('ascii'))` indicates that the read data is decoded using ASCII.

In this example, the code displays the output from S1 on the console within one second after executing the **display cu** command.

Issue other required commands to the device using `write()` in the same way.

(4) Closing the session

```
tn.close()
```

Invoke `close()` to end the current session. The number of VTY connections on the device is limited. Therefore, you need to close the Telnet session after the script is executed.



4 Verification

Omitted.

5 Quiz

1. The port number used by Telnet is ().
 - A. 20
 - B. 21
 - C. 22
 - D. 23
2. The port number used by SSH is ().
 - A. 20
 - B. 21
 - C. 22
 - D. 23
3. In the Telnet class of the telnetlib module, the method used to read all data that does not block I/O is ().
 - A. read_until()
 - B. read_very_eager()
 - C. close()
 - D. write(buffer)